				$N = 500$		$N = 1000$		$N = 2000$		$N = 4000$		Slope	
(α_g, α_ℓ)	Signs	$\gamma'\delta$	Unadj.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Est.	Pred.
A. DML-DiD, post-Lasso													
(0.7, 0.7)	aligned	1.00	0.83	0.511	0.04	0.445	0.00	0.377	0.00	0.320	0.00	0.27	0.21
(2, 2)	aligned	1.00	0.83	0.065	0.93	0.045	0.90	0.018	0.95	0.016	0.93	−0.25	−0.25
(0.8, 1.5)	aligned	0.87	0.72	0.180	0.61	0.121	0.66	0.083	0.66	0.057	0.66	−0.05	−0.02
B. DML-DiD, Lasso (no refit)													
(0.7, 0.7)	aligned	1.00	0.83	0.729	0.00	0.648	0.00	0.566	0.00	0.488	0.00	0.31	0.21
(2, 2)	aligned	1.00	0.83	0.279	0.23	0.181	0.29	0.121	0.31	0.079	0.41	−0.11	−0.25
(0.8, 1.5)	aligned	0.87	0.72	0.402	0.02	0.301	0.00	0.220	0.00	0.161	0.00	0.06	−0.02

Table 1: D2: bias and coverage of DML-DiD by decay cell, aligned signs, over $R = 500$ replications ($K = 5$, $p = N/2$, $\theta_0 = 1$). $\gamma'\delta$ and the bias of the unadjusted DiD (Unadj.) are at $N = 4000$. Slope: estimated slope of $\log(\sqrt{N}\,|\text{bias}|)$ on $\log N$, and the slope $(e_g + e_\ell - 1)/2$ with $e = 1/(2\alpha)$ predicted by the Lasso rates. The Monte Carlo standard error of the bias is at most 0.006.

				$N = 500$		$N = 1000$		$N = 2000$		$N = 4000$		Slope	
(α_g, α_ℓ)	Signs	$\gamma'\delta$	Unadj.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Est.	Pred.
A. DML-DiD, post-Lasso													
(0.7, 0.7)	aligned	1.00	0.83	0.511	0.04	0.445	0.00	0.377	0.00	0.320	0.00	0.27	0.21
(0.7, 0.7)	random	−0.52	−0.43	−0.146	0.78	−0.091	0.80	−0.079	0.72	−0.058	0.73	–	–
(2, 2)	aligned	1.00	0.83	0.065	0.93	0.045	0.90	0.018	0.95	0.016	0.93	−0.25	−0.25
(2, 2)	random	−0.85	−0.71	0.073	0.91	0.036	0.91	0.014	0.94	0.010	0.96	–	–
(0.8, 1.5)	aligned	0.87	0.72	0.180	0.61	0.121	0.66	0.083	0.66	0.057	0.66	−0.05	−0.02
(0.8, 1.5)	random	0.76	0.63	0.095	0.84	0.022	0.93	0.001	0.95	0.000	0.95	–	–
B. DML-DiD, Lasso (no refit)													
(0.7, 0.7)	aligned	1.00	0.83	0.729	0.00	0.648	0.00	0.566	0.00	0.488	0.00	0.31	0.21
(0.7, 0.7)	random	−0.52	−0.43	−0.354	0.17	−0.243	0.18	−0.173	0.15	−0.128	0.10	–	–
(2, 2)	aligned	1.00	0.83	0.279	0.23	0.181	0.29	0.121	0.31	0.079	0.41	−0.11	−0.25
(2, 2)	random	−0.85	−0.71	−0.133	0.72	−0.044	0.87	−0.009	0.91	0.003	0.94	–	–
(0.8, 1.5)	aligned	0.87	0.72	0.402	0.02	0.301	0.00	0.220	0.00	0.161	0.00	0.06	−0.02
(0.8, 1.5)	random	0.76	0.63	0.316	0.15	0.204	0.17	0.123	0.28	0.067	0.49	–	–

Table 2: D2/D3: as Table 1, for aligned and random signs. For random signs, slopes are not reported: the bias either falls below its Monte Carlo error or changes sign as more coordinates are selected, so $\log|\text{bias}|$ does not measure a rate.

			Post-Lasso				Lasso (no refit)			
(α_g, α_ℓ)	Signs	N	Bias	B_1	Formula	Both omit	Bias	B_1	Inner	Both omit
(0.7, 0.7)	aligned	500	0.511	0.528	0.520	0.655	0.729	0.734	0.900	0.630
(0.7, 0.7)	aligned	4000	0.320	0.320	0.317	0.357	0.488	0.486	0.543	0.336
(0.7, 0.7)	random	500	−0.146	−0.138	−0.139	−0.176	−0.354	−0.354	−0.418	−0.171
(0.7, 0.7)	random	4000	−0.058	−0.057	−0.057	−0.063	−0.128	−0.124	−0.147	−0.060
(2, 2)	aligned	500	0.065	0.074	0.070	0.076	0.279	0.280	0.307	0.069
(2, 2)	aligned	4000	0.016	0.014	0.014	0.015	0.079	0.077	0.076	0.014
(2, 2)	random	500	0.073	0.062	0.060	0.067	−0.133	−0.137	−0.152	0.060
(2, 2)	random	4000	0.010	0.008	0.008	0.009	0.003	0.004	0.003	0.009
(0.8, 1.5)	aligned	500	0.180	0.197	0.190	0.240	0.402	0.408	0.498	0.230
(0.8, 1.5)	aligned	4000	0.057	0.055	0.055	0.064	0.161	0.158	0.177	0.060
(0.8, 1.5)	random	500	0.095	0.088	0.091	0.123	0.316	0.311	0.377	0.116
(0.8, 1.5)	random	4000	0.000	0.001	0.002	0.001	0.067	0.069	0.078	0.001

Table 3: D2/D3: the bias against B_1 and the Gaussian bias formula of Proposition 9, at the smallest and largest sample size. B_1: the Lemma 4 term on an independent test draw, with the fitted propensity unclipped. Formula: all three terms of Proposition 9. Both omit: $\bar{c}_2 \sum_{j \notin \hat{S}_g \cup \hat{S}_\ell} \gamma_j \delta_j$. Inner: $\bar{c}_2 \, \Delta\gamma' \Delta\delta$. Averages over $R = 500$ replications.